Brain Tumor Detection Using Deep Learning: A Binary Image Classification Approach

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Abstract

This report presents a deep learning model that classifies brain MRI scans as glioma (tumor) or normal. A ResNet50 network pretrained on ImageNet is fine-tuned on 769 images from the PMRAM Bangladeshi Brain Cancer MRI dataset. An earlier version of this work reported 98% accuracy, but that figure was measured on the same validation images used to select the model. This revision adds a held-out test set, bootstrap confidence intervals, early stopping, and a split that keeps duplicate scans (the 769 images contain only 681 unique scans) out of the test set. On 115 unseen test images the model reaches **95.7% accuracy** (95% CI 91.3 – 99.1%) and a ROC-AUC of 0.989, detecting 51 of 56 tumors with no false alarms. Grad-CAM heatmaps show that the model usually focuses on the tumor, but also reveal off-target attention and a possible shortcut related to MRI sequence type, both discussed as limitations.

1. Introduction

Brain tumors are among the most life-threatening forms of cancer, requiring early and accurate diagnosis to improve patient outcomes. Medical imaging, particularly Magnetic Resonance Imaging (MRI), is a critical tool for brain tumor detection and classification. However, manual analysis of these images by radiologists can be time-consuming, prone to human error, and limited by subjectivity. With the growing availability of annotated medical image datasets and advancements in deep learning, automated brain tumor detection systems have the potential to assist clinicians and improve diagnostic accuracy.

This report presents a deep learning-based binary classification model to detect the presence or absence of brain tumors in MRI images. The model is trained on a curated dataset of brain MRI images labeled as either tumor (glioma) or normal. A convolutional neural network (CNN) learns hierarchical image features and classifies unseen scans, and data augmentation is applied to increase robustness and reduce overfitting. Beyond the classifier itself, this revision focuses on *evaluating it honestly*: measuring performance on data the model has never seen, quantifying the uncertainty of that measurement, and using explainability to check what the model has actually learned.

2. Dataset

The images come from the PMRAM: Bangladeshi Brain Cancer – MRI Dataset [1], published on Mendeley Data under a CC BY 4.0 license. It contains four classes (glioma, meningioma, pituitary tumor and no tumor), each with about 400 raw images, plus a separately augmented copy of the data. This project uses only the **raw** images of two classes: 373 glioma and 396 normal scans, 769 in total. Labels are assigned from the image filenames and stored in a CSV file (1 = glioma, 0 = normal). The pre-augmented images are deliberately not used, because augmented copies of one scan would end up in both training and test data.

2.1 Duplicate scans

While evaluating the first version of the model, exact copies of the same scan were found saved under different filenames (Figure 1). To find them systematically, every image is reduced to a 256-bit perceptual hash (a 16×16 grayscale thumbnail where each pixel is compared with the mean). Pairs whose hashes differ in at most 5 bits are then checked at the pixel level, and are treated as duplicates only if their 64×64 thumbnails differ by at most 4 grey levels on average. The second check is necessary: the hash alone matched different scans with similar dark backgrounds and chained them into one large group of 71 images with mixed labels. With both checks, the 769 images form **681 unique scans**; the largest group holds 4 copies and no group mixes labels.

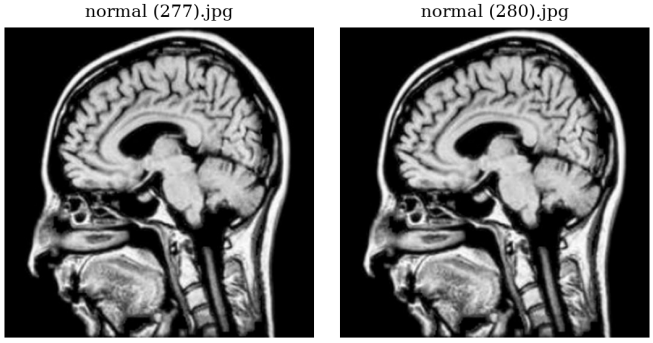


Figure 1. Two files in the dataset that contain the same scan. If one lands in training and the other in testing, the test measures memorization rather than generalization.

2.2 Train, validation and test split

The data is split into training (70%), validation (15%) and test (15%) sets, stratified by class so that each set keeps the overall tumor ratio. Whole duplicate groups are assigned to a single set, so no scan appears in more than one. The resulting sets contain 538, 116 and 115 images. The training set is used to fit the model, the validation set to decide when to stop training and which epoch to keep, and the test set only once, for the final results.

3. Methodology

3.1 Preprocessing and augmentation

Images are converted to RGB, resized to 224×224 pixels and normalized with the ImageNet channel means and standard deviations. During training, each image is randomly flipped horizontally, rotated by up to ±10° and varied in brightness and contrast by up to ±20%. Validation and test images are only resized and normalized.

3.2 Model architecture

The model is a ResNet50 [2] pretrained on ImageNet [3] (transfer learning). The early layers, which detect generic features such as edges and textures, are frozen. Only the last residual block (*layer4*) and a new classification head are trained: a dropout layer ($p = 0.3$) followed by a linear layer with two outputs. This leaves 15.0 million of the 23.5 million parameters trainable (63.7%). The batch normalization layers inside ResNet50 and the dropout in the head help to prevent overfitting.

3.3 Training procedure

The model is trained with cross-entropy loss over the two classes and the Adam optimizer (learning rate 10^{-3}, weight decay 10^{-4}), with a batch size of 32. The learning rate is halved when the validation loss stops improving for two epochs. Training runs for at most 15 epochs, with **early stopping**: it ends once the validation loss has not improved for 5 consecutive epochs. The epoch with the lowest validation loss is kept as the final model. Validation loss is used rather than validation accuracy because, with about 115 validation images, accuracy changes in steps of almost 1% and frequently ties between epochs, while the loss changes smoothly. All random seeds are fixed (seed 42), and repeated runs reproduce identical results.

3.4 Evaluation

The final model is evaluated once on the test set using accuracy, precision, recall, F1-score (macro averages over both classes) and the area under the ROC curve (ROC-AUC). Because the test set is small, each metric is reported with a 95% bootstrap confidence interval [4]: the 115 test predictions are resampled with replacement 1,000 times, the metric is recomputed on each resample, and the 2.5th and 97.5th percentiles form the interval.

4. Results

Training stopped after 7 epochs. The validation loss was lowest at epoch 2 and rose afterwards while training accuracy approached 100% (Figure 2), a sign of overfitting that early stopping and loss-based model selection guard against. The epoch-2 model was therefore evaluated on the test set (Table 1).

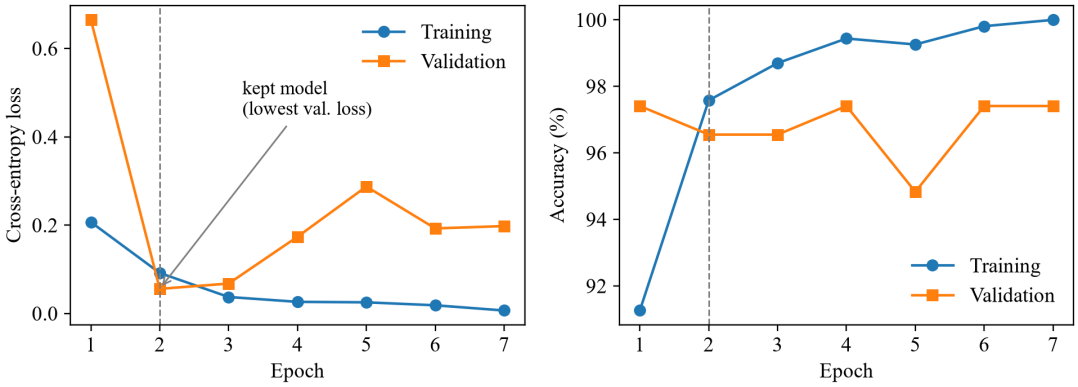


Figure 2. Loss and accuracy per epoch. The dashed line marks epoch 2, the lowest validation loss and the model that was kept; training stopped five epochs later.

Metric (test set, n = 115)	Score	95% confidence interval
Accuracy	95.7%	91.3 – 99.1%
Recall (macro)	95.5%	91.3 – 99.0%
Precision (macro)	96.1%	92.4 – 99.1%
F1-score (macro)	95.6%	91.3 – 99.1%
ROC-AUC	0.989	0.971 – 1.000

Table 1. Results on the held-out test set.

The confusion matrix (Figure 3) shows that every error is a missed tumor: 5 of 56 glioma scans were classified as normal (tumor recall 91.1%), while all 59 normal scans were classified correctly (normal recall 100%). In a screening context missed tumors are the more serious error, so improving tumor recall is the main goal for future work. The width of the confidence intervals (about ±4 percentage points) is a direct consequence of the small test set and should be kept in mind when comparing with other results.

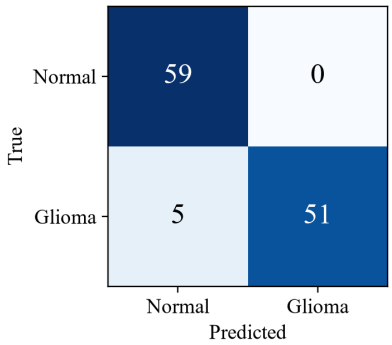


Figure 3. Confusion matrix on the test set.

5. Explainability with Grad-CAM

Grad-CAM [5] highlights the image regions that most influenced a prediction. The gradients of the predicted class score with respect to the feature maps of the last convolutional block are averaged into one weight per feature map; the weighted sum of the maps, with negative values removed, gives a coarse 7×7 heatmap that is upsampled onto the image. The heatmap shows *where the model is looking*. It is not a segmentation of the tumor.

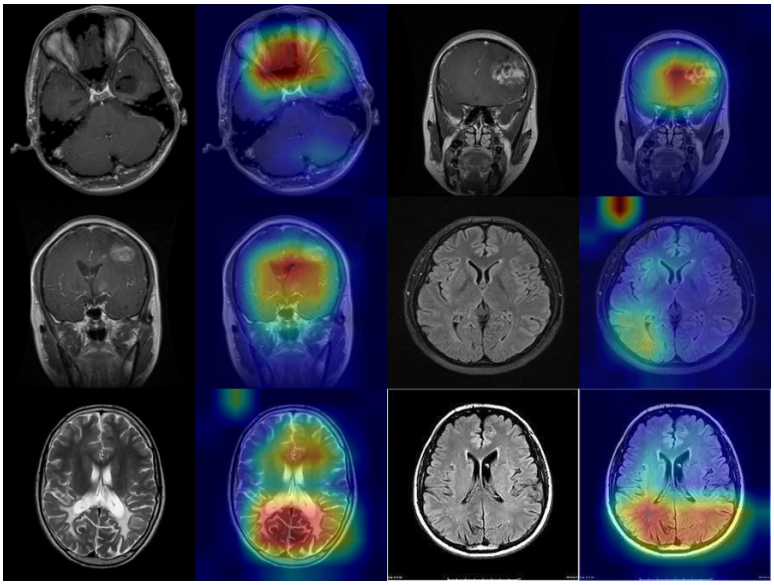


Figure 4. Test-set scans next to their Grad-CAM heatmaps (top row and middle left: glioma; others: normal). All six were classified correctly.

On most glioma scans the heatmap lands on or next to the tumor, which suggests the model has learned relevant features. It does not always: on the first glioma example the attention is in the frontal region rather than on the lesion, and on two normal scans a hot spot appears in an image corner outside the skull, suggesting the model also uses acquisition artifacts. Grad-CAM therefore serves as a check on *why* the model is right, not only *whether* it is.

6. Discussion

6.1 Comparison with the earlier result

The first version of this project reported 98% accuracy and recall. That figure was measured on the validation set that was also used to choose the best epoch, so it was optimistic. Table 2 shows how the result changed as the evaluation was corrected. The test set was consulted once after each of the first two changes; each change was motivated by a problem visible without the test results (ties in validation accuracy, and duplicate files), and no further changes were made after the final result.

Evaluation setup	Test accuracy
Original: validation set doubles as test set	98% (validation)
Held-out test set, model chosen by best validation accuracy	94.0%
Model chosen by lowest validation loss (duplicates not yet handled)	95.7%
Duplicate-aware split (final, this report)	95.7%

Table 2. Evaluation history. Model selection by validation accuracy kept the barely-trained first epoch, because later epochs only tied its accuracy.

6.2 Limitations

- **Single dataset.** The model was trained and tested on images from one source; its performance on scans from other hospitals, scanners or protocols is unknown.
- **Same-patient slices.** The dataset provides no patient identifiers. Exact and near-duplicate images are kept within one split, but different slices of the same patient (for example consecutive slices through one tumor) can still appear in both training and test data, which may make the test score optimistic.
- **Possible shortcut.** Most glioma images are T1-weighted post-contrast scans, while most normal images are T2-weighted or FLAIR scans. The model may partly recognize the MRI sequence rather than the tumor. A fair test would compare tumor and normal scans of the same sequence.
- **Small test set.** With 115 test images, the 95% confidence interval for accuracy spans about 8 percentage points.
- **Binary task.** The model does not distinguish tumor types (glioma, meningioma, pituitary) or grades, which are crucial in clinical practice.

7. Demo Application and Reproducibility

A Gradio web application lets users upload an MRI scan, or choose one of six sample scans from the test set, and returns the prediction, the class probabilities and the Grad-CAM overlay (Figure 5). The application states clearly that it is a research demo and not a medical device. The source code, automated tests and trained model weights are available on GitHub:

github.com/musawaghu/Brain-Tumor-Detection-DL

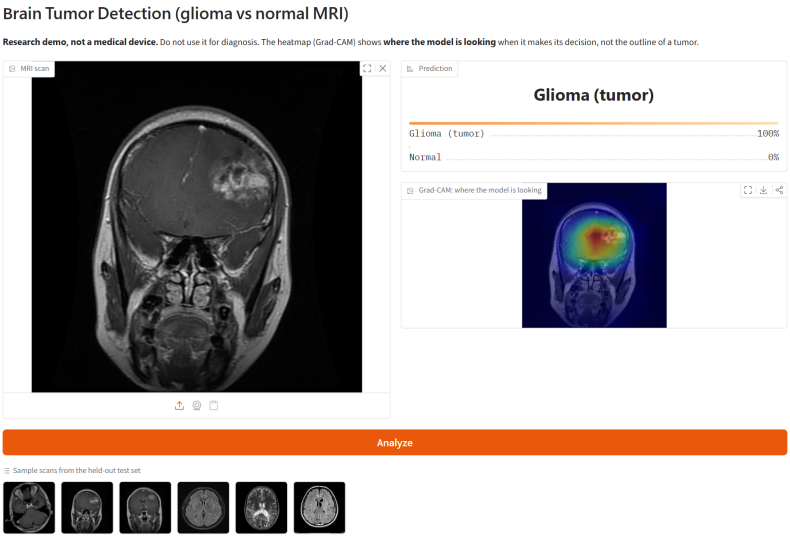


Figure 5. The demo application on a glioma test scan.

8. Conclusion and Future Work

This study presents a deep learning approach for automated brain tumor detection using binary classification of MRI images. A fine-tuned ResNet50 reaches 95.7% accuracy (95% CI 91.3 – 99.1%) and a ROC-AUC of 0.989 on a held-out test set from which duplicate scans were excluded. The revision shows that how a model is evaluated matters as much as the model itself: reusing validation data, selecting epochs on a noisy metric and duplicate files each changed the reported accuracy by several percentage points.

Future work will focus on improving tumor recall, testing on data from other sources, controlling for MRI sequence type, expanding the model to multi-class classification (glioma, meningioma, pituitary and normal), and, with suitable annotations, moving from heatmaps to tumor segmentation. Combined with careful validation, such models could support radiologists, especially in regions with limited access to expert readers.

References

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